ECEN 248 - Lab Report

Lab Number: 2

Lab Title: Logic Minimization and Karnaugh Maps

Section Number: 509

Student's Name: Saira Khan

Date: 9/21/2022

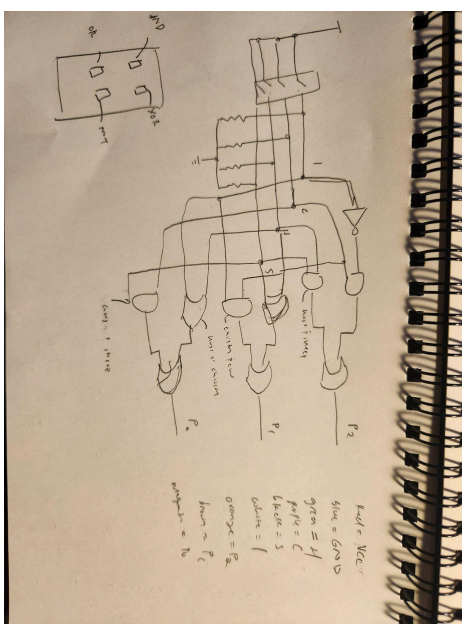
TA: Sri Kodi

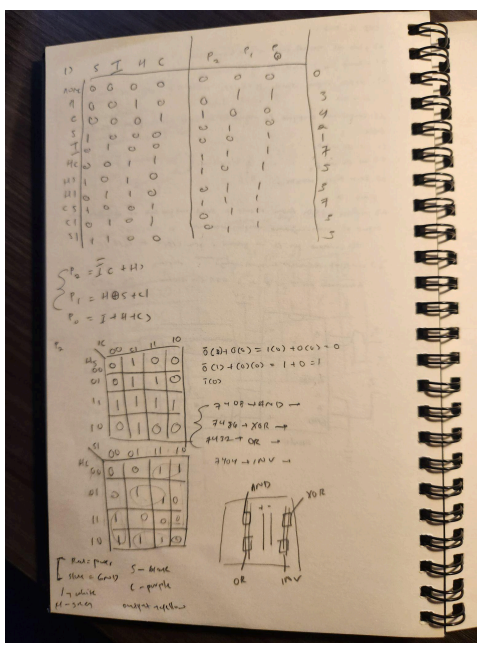
Objectives:

The purpose of this lab is to learn about Karnaugh-Maps and their use of logic minimization. The lab will demonstrate how to use breadboards to create a circuit based on truth tables and circuit drawings as well as how to use a seven-segment digit display. the purpose of the lab is to simplify the circuits built in order to make troubleshooting easier.

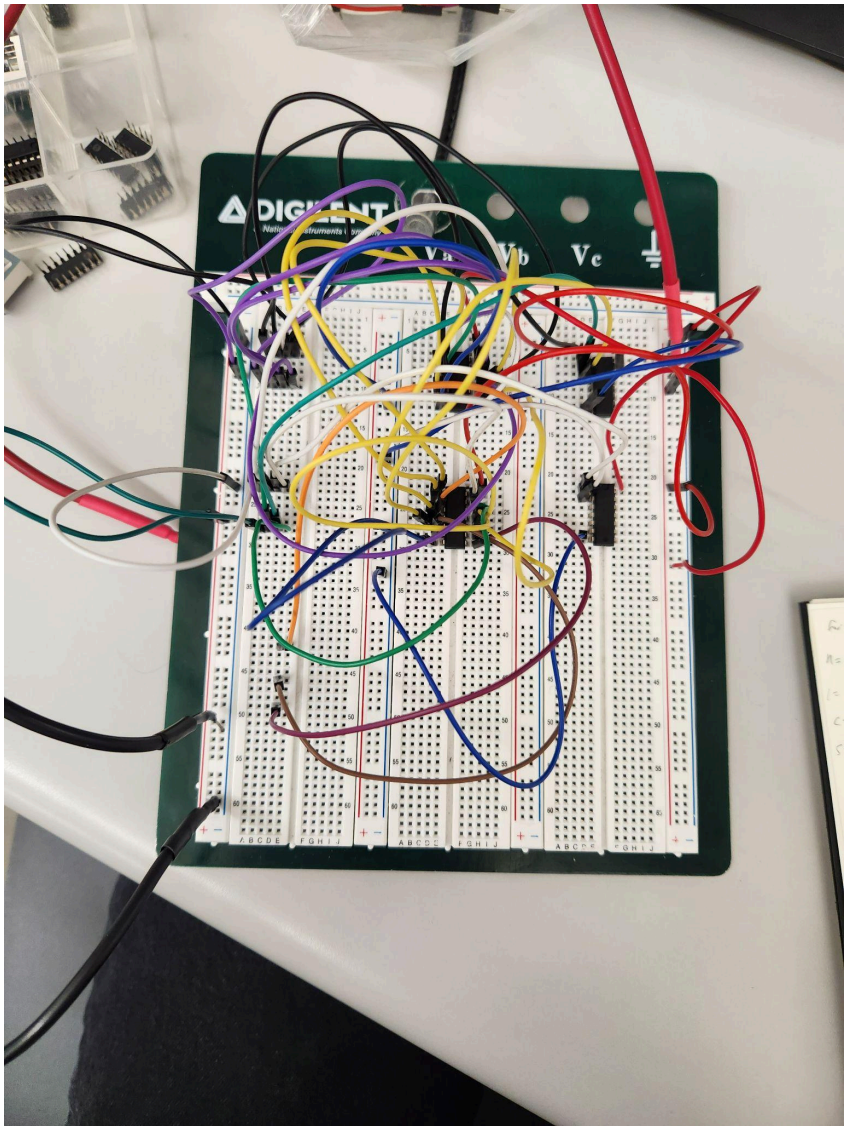
Design:

In order to complete this lab, the first step was creating a truth table, a Karnaugh map, and a circuit diagram in line with the rules given. The circuit diagram was then used to set up the breadboard. An XOR, NOT, AND, and OR gate were used and their supporting ICs were placed on the breadboard. From there, power and ground were connected to each logic gate, and circuits were used to correspond to the diagram to set up the calculator. Certain colored wires were used for the sheep, cows, hogs, chickens, outputs, P0, P1, and P2. They were connected accordingly and then the circuit was tested using the truth table made before. Switching some of the inputs from 0 to 1, the circuit was tested to ensure that the output was correct. Once the circuit was tested for all cases and proven to work, a decoder was put on the breadboard and the outputs were fed to it. Then the rest of the decoder was assembled by connecting the right wires to a seven-segment digit display, where two resistors were used. Once the decoder was set up correctly, the output was tested again on the seven-segment digit display.





Truth table, Karnaugh Map, and Circuit used to build the circuit



Circuit for Part 1

Results:

The results were as expected from the truth table. For values outputted as ‘0’, the DC power supply outputted a voltage usually less than 0 V or around 1 V. While with values of ‘1’, the DC power supply outputted a voltage around 4 V.

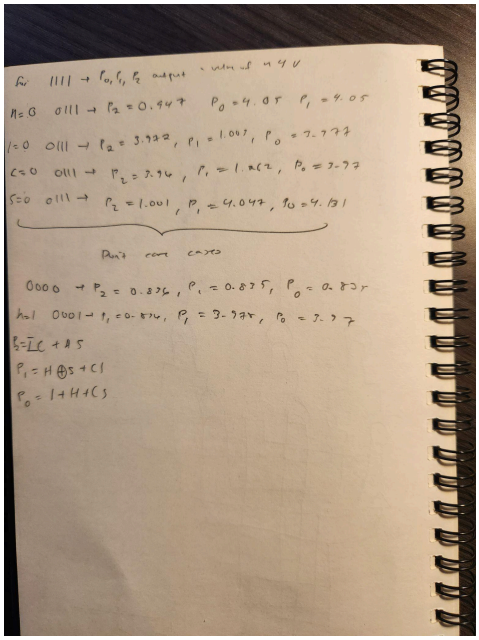
	S	I	H	C	P2	P1	P0
none	0	0	0	0	0	0	0
H	0	0	1	0	0	1	1

C	0	0	0	1	1	0	0
S	1	0	0	0	0	1	0
I	0	1	0	0	0	0	1
HC	0	0	1	1	1	1	1
HS	1	0	1	0	1	0	1
HI	0	1	1	0	0	1	1
CS	1	0	0	1	1	1	1
CI	0	1	0	1	0	1	1
SI	1	1	0	0	0	1	1
HICS	1	1	1	1	1	0	1
ICS	1	1	0	1	0	1	1
HIS	1	1	1	0	1	0	1
CHI	0	1	1	1	0	0	1
HSC	1	0	1	1	1	0	1

“Don’t Care” Values in Voltage

	H	C	S	I	P2	P1	P0
HSCI	1	1	1	1	4 V	4.11	4.02 V
CSI	0	1	1	1	0.947 V	4.05 V	4.05 V
HCS	1	1	1	0	3.972 V	1.003 V	3.977 V
HIS	1	0	1	1	3.96 V	1.262 V	3.97 V
HCI	1	1	0	1	1.001 V	4.047 V	4.131 V

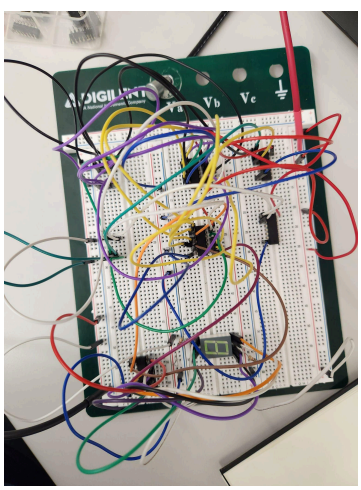
The values that the circuit outputs for the “don’t care” values are represented in the table above with voltage outputted by the DC power supply. ‘0’s were represented by voltage from 0 to around 1 volt. While ‘1’ was outputted to be about 4 volts.



Conclusion:

Through the lab, I learned how to use a Karnuagh Map to simplify logic circuits as well as how to feed output to a seven-segment display. I did this by creating a truth table, a Karnuagh map, and a circuit diagram and used those to build a circuit on a breadboard. I was then able to test the outputs and compare them with my truth table to see if my circuit was working correctly.

Post-lab Deliverables:



Seven-Segment Display

As shown in the picture, the red wires are for power, the blue wires are for ground, the green is for hogs, the purple is for cows, the black for sheep, the white for chicken, the orange for P2,

the yellow for output of logic gates, brown for P1, and magenta for P0. The logic gates are implemented according to the circuit diagram, and the outputs are fed to a decoder. The decoder then has wires that go to a seven-segment display with two resistors. In order to change the input, wires on the very left of the breadboard are moved according to what animals John wants to use. So if John wanted cows on his farm, he'd move the purple wire from ground to power (0 to 1).